the output vector is copied and fed back into the layer process as part of the input, hence the recurrence.

String diagrams are a very general type of diagram; in addition to diagramming neural networks, we could use them to diagram how to bake a cake. A *computational graph* is a special kind of string diagram where all the processes represent concrete computations that a computer can perform, or that can be described in some programming language like Python. If you've ever visualized a computational graph in TensorFlow's TensorBoard, you'll know what we mean. The goal of a good string diagram is that we can view an algorithm or machine learning model at a high level to get the big picture, and then gradually zoom in until our string diagram is detailed enough for us to actually implement the algorithm based almost solely on our knowledge of the diagram.

Between the mathematics, simple Python code, and string diagrams that we'll present in this book, you should have no problem understanding how to implement some pretty advanced machine learning models.

2.2. Solving the multi-arm bandit

We're now ready to get started with a real reinforcement learning problem and look at the relevant concepts and skills needed to solve this problem as we go. But before we get too fancy, building something like AlphaGo, let's first consider a simple problem. Let's say you're at a casino, and in front of you are 10 slot machines with a flashy sign that says "Play for free! Max payout is $10!" Wow, not bad! Intrigued, you ask one of the employees what's going on, because it seems too good to be true, and she says, "It's really true, play as much as you want, it's free. Each slot machine is guaranteed to give you a reward between $0 and $10. Oh, by the way, keep this to yourself, but those 10 slot machines each have a different average payout, so try to figure out which one gives the most rewards on average, and you'll be making tons of cash!"

What kind of casino is this? Who cares, let's just figure out how to make the most money! Oh by the way, here's a joke: What's another name for a slot